MILITARY SPECIFICATION

TELESCOPE, PANORAMIC: M115

1. SCOPE

1.1 This specification covers one type of telescope known as Telescope, Panoramic: M115 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U.S. Army, Frankford Arsenal

F8587340	Telescope, Panoramic: Ml15

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PACKAGING DATA SHEET

8587340 Packaging of Telescope, Panoramic: M115

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. RETIREMENTS

3.1 Fabrication.- The telescope shall be manufactured in accordance with "Drawing F8587340 and drawings pertaining thereto.

3.2 General specifications.- The contractor shall be responsible for adherence to, and compliance with the following requirements of MIL-O-13830 and MIL-F-13926.

3.2.1 MIL-O-13830.

Illuminated reticle

3.2.2 MIL-F-13926.

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Electrical and electronic assemblies
(e) Part identification and marking
(f) Workmanship

3.3 Environmental.

3.3.1 Storage temperature.- The telescope shall show no evidence of physical failure and shall meet all requirements of this specification at standard ambient temperature (plus 60 degrees to 90 degrees Fahrenheit) after having been exposed and thermally stabilized at ambient temperatures of plus 160 degrees and minus 80 degrees Fahrenheit.

3.3.2 Operating temperature.- The telescope shall meet the applicable requirements of 3.9 to 3.9.2.2 inclusive while exposed and thermally stabilized at ambient temperatures of plus 150 degrees and minus 40 degrees Fahrenheit. Upon return to standard ambient temperature, the telescope shall meet the requirements of 3.3.3.

3.3.3 Vibration.- The" telescope shall be vibrated in a vertical plane at a constant frequency of thirty cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of five minutes plus or minus fifteen seconds. Subsequent to vibration there shall be no evidence of physical failure and the telescope shall meet the requirements of 3.4 to 3.11 inclusive. (see 3.7.1.1)

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3.4 Sealing and purging.

3.4.1 Optical assembly area.- The telescope optical assembly area shall be capable of sustaining an internal pressure of 5.0 ± 0.1 psi for a period of 2 hours minimum without loss of pressure in excess of 0.2 psi.

3.4.2 Counterbox area.- The counterbox area of the telescope shall be capable of sustaining an internal pressure of 2.0 ± .2 psi for a period of 2 hours minimum without loss of pressure in excess of 0.2 psi.

3.4.3 Purging.- The optical assembly portion of the telescope shall be purged with dry nitrogen supplied at a pressure of 5 ± 1/4 psi until the dew point of the emergent gas is no higher than -25°F. Purging apertures shall be quickly and positively closed to minimize intermingling of external atmosphere upon disconnection of purging equipment.

3.5 Cleanliness.

3.5.1 Dirt.- Inspection for dirt and foreign particles (classed as dirt) shall be made from the eye end only. The maximum size of any particle appearing on the reticle shall be a reticle line width. The maximum number of dirt particles appearing within a central field of view of 50 mils diameter shall be three (3). The maximum number of dirt particles appearing outside this field of view shall be three (3). The minimum angular separation between dirt particles shall be 15 mils. Dirt particles smaller than 1/4 reticle line width shall be ignored regardless of distribution.

3.5.2 Other defects.- There shall be no evidence of moisture, grease, fingerprints, condensates, fractures, and adhesive separations when viewing through the objective and eye end of the instrument.

3.6 Orientation.- The requirements of 3.7 to 3.8.3.3.2, inclusive, shall be met with the flat mounting surface of the telescope horizontal, the elbow assembly within 2° of the center of the swing stops, the head elevation indices indicating zero, the azimuth and reset counters indicating 3200 mils, the gunner's aid counters set at zero, and the five mil click lead knob set for "Indirect" fire (out of detent).

3.7 Accuracy.

3.7.1 Collimation.

3.7.1.1 Pre-vibration setting.- Prior to meeting the requirements of 3.3.3, the line of sight shall be adjusted horizontal and perpendicular to the locating keyways within 0.25 mil.

3.7.1.2 Post-vibration.- Subsequent to meeting the requirements of 3.3.3, the line of sight shall be within 0.25 mil of the position established in 3.7.1.1 excluding backlash.

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3.7.1.3 Open sight.- With the telescope positioned as in 3.6, the line of sight of the open sight shall be parallel in azimuth and elevation to the line of sight through the telescope optics within 10 mils.

3.7.2 Image tilt.- The image of a vertical target line shall be vertical within one degree of arc throughout two (2) revolutions of the head.

3.7.3 Image wander.- The wander of the reticle image measured at the eyepiece shall not exceed two degrees in diameter for two (2) complete revolutions of the head assembly, In addition, all reticle graduations shall be visible throughout the revolutions of the head. This requirement shall also be met with the elbow assembly positioned against each of the swing stops.

3.7.4 Parallelism of reticle and image.- The vertical reticle line shall be parallel to the image of a vertical target line within 30 minutes of arc at the position established in paragraph 3.7.1.1.

3.7.5 Eyepiece focus.- The best eyepiece focus of the reticle image shall be fixed between the limits of -0.75 and -1.00 diopters. The variation in eyepiece focus between the horizontal and vertical reticle line at the center of the reticle image shall not exceed 1/4 diopter.

3.7.6 Parallax.- Parallax at the center of the field shall not exceed 0.15 mil when viewing a target at 130 meters ± 10 meters.

3.7.7 Resolution.- Resolution on the optical axis shall be 15 seconds of arc or less when using an observation telescope of at least 3 power and at least 0.4 inch entrance pupil. resolution consists of detactable line structure and proper line count in all four meridians. Total spread in focus shift between meridians shall not exceed 1/4 diopter measured at the instrument eyepiece.

3.7.8 Reticle.- The angular separation between the two 40 mil graduations on the horizontal axis shall be 80 mils plus or minus 0.5 mil.

3.8 Performance.

3.8.1 Elevation mechanism.

3.8.1.1 Excursion range.- Rotation of the elevation knob shall cause the line of sight to elevate and depress a minimum of 300 mils from the “position specified in 3.7.1.1.

3.8.1.2 Plumb travel.- Starting with the line of sight in coincidence with a vertical target line at zero elevation,the line of sight shall track the target line within 0.5 mil (total spread of 1.0 mil) when elevated 300 mils, depressed to 300 mils depression, then elevated to zero elevation.

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3.8.1.3 Backlash.- Backlash in the elevation mechanism shall not exceed 5.0 mils.

3.8.2 Azimuth mechanism.

3.8.2.1 800 Mil steps.- Error at deflection readings in 800 mil steps throughout two full revolution of the head in one direction shall not exceed 1.0 mil including backlash. The total spread between the greatest positive error and the greatest negative error shall not exceed 1.0 mil. The total error, excluding backlash, between the 6400 mil azimuth counter reading and the reset counter for any one number shall not exceed 0.25 mil.

3.8.2.2 15 Mil steps.- Rotating the gunner's aid knob to indicate 15 mils, followed by rotation of the azimuth knob in the opposite direction, with backlash removed, to deflect the line of sight 15 mils, shall cause the reset counter to return to within 0.25 mil of the original setting. This requirement shall be met with the 15 mil value set into each gunner's aid counter at azimuth positions 3200, 4800, zero, and 1600 mils as read on the azimuth counter.

3.8.2.3 Backlash.- Backlash in the azimuth mechanism shall not exceed 0.5 mil when read on the azimuth counter, or 0.75 mil when read on the reset counter.

3.8.2.4 Level travel.- Starting with the line of sight in coincidence with a horizontal target, the line of sight shall track the target within 1.0 mil (total spread of 2 mils) when the head is rotated through two full revolutions in azimuth.

3.8.2.5 Lift.- Vertical displacement of the line of sight due to reversing rotation by means of the azimuth knob shall not exceed 0.5 mil.

3.8.2.6 Five mil click lead mechanism.- With the lead knob set for "Direct" fire, a detent action shall be felt and heard when the azimuth knob is rotated to introduce increments of 5.0 mils ± 0.5 mil into the 6400 mil azimuth counter. This requirement shall be met at standard and extreme operating temperatures.

3.8.3 Counter mechanisms.

3.8.3.1 Rapid traverse.- The reset counter shall follow the azimuth counter within 0.25 mil when the azimuth knob is rotated at 150 rpm for one minute. The gunner's aid counters shall remain at zero.

3.8.3.2 Reset counter.- The reset counter shall be resettable to 3200 mils ± 0.25 mil from any counter reading.

3.8.3.3 Gunner's aid counters.

3.8.3.3.1 Excursion range.- The gunner's aid counters designated by "L" and "R" shall each have a minimum excursion Of 50 milS.

3.8.3.3.2 Settings.- Setting 50 mils on the "L" counter shall decrease the reset counter reading by 50 mils ± 0.5 mil. Setting 50 mils on the "R" counter shall increase the reset counter reading by 50 mils ± 0.5 mil.

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There shall be no change greater than .25 mil in the azimuth counter readings and no movement of the line of sight when the 50 mil values are set into the gunner's aid counters.

3.9 Operability.- The boresight adjustment shaft and the azimuth counter door and latch shall be operable at the extreme operating temperatures of 3.3.2.

3.9.1 Torque.- The running torque required to operate the following knobs shall be within the values specified for standard ambient temperature of +60° to +90°F, and shall not exceed the values specified for extreme operating temperatures. The torque values for the azimuth knob shall be met with the five mil click lead knob set for "Indirect" fire (out of detent).

KNOB	AT +60°F TO +90°F	AT -40°F AND +150°F
Azimuth	3 to 8 in. lbs.	15 in. lbs. max.
Gunner's aid	10 to 45 in. ozs.	60 in. ozs. max.
Reset counter	36 in. ozs. max.	46 in. ozs. max.
Elevation	5 to 40 in. ozs.	80 in. ozs. max.

3.9.2 Direct force.- The following operability requirements shall be met with the force applied horizontally and in the direction of movement.

3.9.2.1 Elbow assembly.- At a standard ambient temperature of +60° to +90°F, a direct force of 2 to 8 pounds, applied adjacent to the eyelens, shall be required to rotate the elbow assembly in each direction through the full extent of travel. The force required at the extreme operating temperatures of 3.3.2 shall not exceed 15 pounds.

3.9.2.2 Reset counter knob.- At a standard ambient temperature of +60° to +90°F, a direct force of 11 pounds maximum shall be required to push in the reset counter knob sufficiently to permit rotation. The force required at the extreme operating temperatures of 3.3.2 shall not exceed 16 pounds.

3.10 Illumination.

3.10.1 Counters.- With the lamp energized, the numerals and markings of each counter shall be clearly distinguishable when observed in a darkened area.

3.10.2 Reticle.- With a 28 volts dc, 0.04 amperes light source centrally positioned 3/8 inch ± 1/32 inch from the illumination slot, the reticle markings shall be clearly distinguishable when observed in a darkened area. This requirement shall be met through one full revolution of the head and when the elbow is rotated through its full range of travel.

3.11 Insulation resistance.- Insulation resistance shall not be less than 10 megohms when a potential of 100 volts direct current is applied between insulated points and ground.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilites or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Goverment. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) panoramic telescopes and three (3) each of every component and subassembly covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor’s inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with “Submission of product” provisions of MIL-STD-105.

4.3.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926, and the Supplementary Quality Assurance Provisions (SQAP) listed

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in the List of Inspection Documents (IEL) when provided with the technical data package (TDP). Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105, and the sampling plans specified in Tables I and II. Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Table I shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE I - CLASSIFICATION OF DEFECTS

CRITICAL: 100% INSPECTION	REQUIREMENT	TEST PROCEDURE
1. Sealing (Counterbox area)	3.4.2	4.6.4.1
MAJOR: AQL 1.0 percent defective		
101. Physical failure, post vibration	3.3.3	4.6.3
102. Sealing (optical assembly)	3.4.1	4.6.4
103. Cleanliness	3.5	4.6.5
104. Collimation	3.7.1.1 to 3.7.1.3 incl.	4.6.7.1 to 4.6.7.3
105. Image tilt	3.7.2	4.6.8
106. Image wander	3.7.3	4.6.8.1
107. Parallelism of reticle and image	3.7.4	4.6.9
108. Eyepiece focus	3.7.5	4,6.10
109. Parallax	3.7.6	4.6.11
110. Resolution	3.7.7	4.6.12
111. Reticle	3.7.8	4.6.13
112. Excursion (elevation)	3.8.1.1	4.6.14
113. Plumb travel	3.8.1.2	4.6.15
114. Backlash (elevation and azimuth mechanisms)	3.8.1.3 and 3.8.2.3	4.6.16 and 4.6.19
115. 800 mil steps	3.8.2.1	4.6.17
116. 15 mil steps	3.8.2.2	4.6.18
117. Level travel (horizontal)	3.8.2.4	4.6.17.1
118. Lift	3.8.2.5	4.6.20
119. Five mil click mechanism	3.8.2.6	4.6.17.2
120. Rapid traverse	3.8.3.1	4.6.21
121. Reset counter (3200 mil)	3.8.3.2	4.6.22
122. Excursion range (gunner's aid counter)	3.8.3.3.1	4.6.23.1
123. Settings (gunner's aid counter)	3.8.3.3.2	4.6.23.2
124. Torque	3.9.1	4.6.24.1
125. Elbow assembly	3.9.2.1	4.6.25.1
126. Reset counter knob	3.9.2.2	4.6.25.2
127. Illumination, counters	3.10.1	4.6.26
128. Illumination, reticle	3.10.2	4.6.26.1
129. Insulation resistance	3.11	4.6.27
MINOR: NONE		

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4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105,

4.4 Environmental.- Three (3) telescopes shall be selected at random from each 50 producted or from each month's production, whichever occurs first, for the following tests specified in Table II.

TABLE II

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
301. Storage temperature	3.3.1	4.6.1
302. Operating temperature	3.3.2	4.6.2
303. Sealing (optical assembly)	3.4.1	4.6.4

4.4.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specification or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents (IEL), when provided with the Technical Data Package (TDP).

4.5.2.2 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

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4.6 Test methods and procedures.

4.6.1 Storage temperatures.- The telescope shall be exposed to the temperatures specified in 3.3.1, utilizing test equipment in accordance with the "Test Facilities" requirements of Specification MIL-F-13926. This equipment shall be capable of maintaining the specified temperatures until thermal stabilization is reached. The temperature shall be changed gradually to avoid thermal shock. After the telescope has been exposed to each extreme temperature and returned to standard ambient temperature, the telescope shall be inspected for evidence of damage or deterioration. The telescope shall then be subjected to examination and the tests herein to determine compliance with all accuracy requirements of this specification.

4.6.2 Operating temperatures.- The telescope shall be subjected to the temperatures specified in 3.3.2, utilizing the test facilities referenced in 4.5.2.2. The temperature shall be changed gradually to avoid thermal shock. When the telescope has been thermally stabilized at each extreme temperature it shall meet the requirements of 3.9 to 3.9.2.2, inclusive. Upon return to standard amibent temperature, the telescope shall be tested in accordance with all remaining tests specified herein to determine compliance with respective requirements referenced in each test.

4.6.3 Vibration.- The telescope shall be mounted on a vibration machine equipped with facilities to securely hold the telescope while the machine is in motion. The telescope shall be vibrated in accordance with 3.3.3 at the amplitude and for the period of time specified. Subsequent to vibration, a visual examination shall be made to determine evidence of physical failure or damage and the telescope shall then meet all requirements as defined in 3.3.3.

4.6.4 Sealing (optical assembly area).- The optical assembly portion of the telescope shall be pressurized internally to the psi value specified in 3.4.1 with dry nitrogen for the length of time specified in 3.4.1. While under pressure, the optical assembly portion shall meet the requirement specified in 3.4.1. Leaks can be determined by using a pressure gage graduated in tenths of a pound (psig) or by using Pressure Testing Fixture, Two Station - 8665556.

4.6.4.1 Sealing (counter box area).- The counter box portion of the telescope shall be pressurized internally to the psi value specified in 3.4.2 with dry nitrogen for the length of time specified in 3.4.2. While under pressure, the counter box portion shall meet the requirement specified in 3.4.2. Dry nitrogen used in this test to have a dew point of at least -25°F. Accuracy of the test equipment used for this test shall be equal to or exceed the accuracies depicted on Pressure Tester Drawing F8665556.

NOTE: This test to be performed on each telescope.

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4.6.4.2 Purging.- The optical assembly area of the telescope shall be flushed with dry nitrogen gas to conform to the requirement of 3.4.3. A dew point tester (commercially available standard equipment) shall be attached to the telescope. Gas emerging from the scope hole in the rotating head of the telescope shall be bled through the tester until it conforms to the value specified in 3.4.3. The telescope shall then be quickly sealed. Pressure of the gas remaining in the telescope shall be brought up to 0.1 to 0.25 psi. Purging shall be conducted at a temperature of +60°F to +90°F.

4.6.5 Cleanliness.- The assembled optical components shall be inspected for dirt by viewing a uniformly illuminated field having a brightness of 300 ± 20 apparent foot candles. Inspection for all other defects shall be made by the technique of shadowing. All defects shall lie within the limits specified in 3.5.

4.6.6 Orientation.- The telescope shall be mounted on Azimuth Test Fixture, Drawing F7691596, by means of an adapter provided with appropriate locating surfaces to receive and position the telescope in the fixture, with the mounting surface, elbow assembly, indices, counters, and lead knob oriented as prescribed in 3.6.

4.6.7 Collimation.

4.6.7.1 Collimation, pre-vibration. - The telescope shall be mounted and set as in 4.6.6 with the line of sight, established by the intersection of the reticle cross lines, in coincidence with a point on the target (or collimator target reticle). Record the position of the telescope reticle on the target. Record the azimuth position on the vernier scale of the counter. These points or settings shall be within the tolerance specified in 3.7.1.1.

4.6.7.2 Collimation, post vibration.- Subsequent to the vibration test of 4.6.3, as required by 3.3.3, the telescope shall meet the requirements of 3.7.1.2 after the telescope has been remounted on the fixture specified in 4.6.6. The line of sight shall not have changed by more than the tolerance specified in 3.7.1.2.

4.6.7.3 Open sight.- The line of sight shall be set to zero elevation (Horizontal). A wall target shall be utilized showing individual targets that have been laid out, one for the line of sight through the telescope optics and one for the line of sight through the open sight. The line of sight through the open sight of the telescope shall be parallel in deflection to the optical line of sight within the tolerance specified in 3.7.1.3.

4.6.8 Image tilt.- The telescope shall be oriented in accordance with 4.6.6. Using an adjustable auxiliary telescope and a flat surface to position the auxiliary telescope at the eyepiece of the telescope under test, adjust the auxiliary telescope to the height of the telescope eyepiece. Adjust the auxiliary telescope's vertical reticle line in coincidence with a plumb line. Sight through the auxiliary telescope and the eyepiece of the telescope to the target. The vertical target line image shall be parallel to the vertical reticle line of the auxiliary telescope within the tolerance specified in 3.7.2. The tolerance can be measured by rotating the deflection (azimuth) knob and reading the error on the 3200 counter vernier scale. The auxiliary telescope's vertical reticle line shall be used as an index.

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4.6.8.1 Image wander.- This test can be performed concurrently with the test specified above. The auxiliary telescope, positioned at the eyepiece, shall contain a calibrated reticle capable of measuring reticle image wander within the limits specified in 3.7.3. Sight through the auxiliary telescope and note the position of the reticle image. Rotate the head assembly through 2 complete revolutions and note the position of the reticle image. The position of the image shall not have deviated from the position first noted by more than the tolerance specified in 3.7.3. This examination shall be repeated with the elbow positioned against the right and left swing stops.

4.6.9 Parallelism of reticle and image.- This test can be performed concurrently with the test specified in 4.6.8. The vertical reticle line of the telescope shall not deviate from the image of the vertical target line by more than the tolerance specified in 3.7.4. The vertical reticle line of the auxiliary telescope shall again be used as the index to measure the error in the telescope. This examination shall also be repeated with the elbow positioned against the right and left swing stops through at least one full revolution of the telescope head in azimuth.

4.6.10 Eyepiece focus.- The telescope eyepiece is a fixed focus assembly and it shall be set between the tolerances specified in 3.7.5. To determine the correct setting of the eyepiece, a dioptometer of three power (3X) shall be used that has been adjusted to the individual eye. This can be accomplished by sighting through the dioptometer at a white background and adjusting the eyepiece of the dioptometer until the reticle is at the sharpest point obtainable. Position the barrel of thr dioptometer to set the index at zero. When this has been accoplished, sight through the dioptometer and the eyepiece of the telescope. Observe the reticle of the telescope, and compare it with the reticle of the dioptometer. Adjust the barrel of the dioptometer to make the reticle of the telescope go out of focus; then re-adjust the barrel to bring the telescope reticle into focus at the sharpest point. Read the error on the scale on the barrel of the dioptometer. Several repeat readings in both directions should be made to determine conformance to requirements.

4.6.11 Parallax.- Parallax shall be determined by viewing a target at the distance specified in 3.7.6. Parallax can be distinguished by adjusting the vertical and horizontal lines of the telescope reticle parallel to the image of the target lines without touching, leaving a minimum of space or light between the lines. The observer shall move his head from side to side or up and down. Any movement of the target in either direction (the space getting wider or narrower) is considered parallax. The amount of parallax observed shall not exceed the tolerance requirements of 3.7.6.

4.6.12 Resolution.- This test shall be performed in accordance with the resolution test specified in MIL-O-13830. The telescope shall permit resolution of a target within the limit specified in 3.7.7.

4.6.13 Reticle.- The test for reticle graduation accuracy shall be conducted in accordance with the reticle requirements in MIL-O-13830 to determine conformance with the requirements of 3.7.8.

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4.6.14 Excursion range (elevation mechanism).- With the line of sight established in accordance with 3.7.1.1, elevation and depression of the prism in the rotating head by means of the elevation knob shall cause the line of sight to displace the amount specified in 3.8.1.1 for elevation and depression, This test can be accomplished by using a target collimator (or a wall target) that has an established level line of sight and the excursion limits required.

4.6.15 Plumb travel.- This test can be performed concurrently with 4.6.14. The telescope shall be in a level position with the vertical reticle line brought into coincidence with the vertical target line by rotating the azimuth knob. The horizontal reticle line of the telescope shall be in coincidence with the horizontal target line by rotating the elevation knob. Rotate the elevation knob to elevate the line of sight of the telescope 300 mils. Reserve the rotation of the knob to depress the line of sight to 300 mils depression. Reverse the rotation again, returning the line of sight to zero elevation. The vertical reticle line of the telescope must track the target line within the tolerance specified in 3.8.1.2.

4.6.16 Backlash (elevation) .- The test for backlash in the elevation mechanism shall be accomplished concurrently with the tests indicated in 4.6.14 and 4.6.15. The horizontal reticle line of the telescope shall be brought into coincidence with the horizontal target line, at zero elevation without overtravel, when the elevation knob is rotated in a clockwise direction. The zero line on the knob and the index line on the rotating head shall be set in alignment with each other, Rotate the elevation knob clockwise to 300 mils, then reverse direction of the knob and align the zero line on the knob with the index line on the head without overtravel. Observe, through the telescope, the position of the telescope reticle on the target. The telescope reticle shall be within the tolerance specified in 3,8.1.3 of the zero level line on the target. This test shall also be conducted in the counterclockwise direction.

4.6.17 800 Mils steps.- Position the test equipment to zero azimuth. Set the telescope azimuth and reset counters to 3200 (zero) and the gunner's aid counters to zero. The vertical line on the reticle of the telescope shall be coincident with the vertical line on the target, and the horizontal line of the reticle shall be coincident with the zero elevation line. Azimuth readings shall be made at 800 mil increments through two full revolutions in each direction. The error in the readings taken shall not exceed the tolerance specified in 3.8.2.1 including backlash.

4.6.17.1 Level travel.- This test shall be accomplished with the test procedure of 4.6.17. During the azimuth reading for each 800 mil increment, the position of the horizontal reticle line of the telescope shall be observed with respect to the zero elevation line on the target. The telescope reticle shall track the zero elevation line on the target within the tolerance specified in 3.8.2.4 for at least one full revolution in azimuth.

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4.6.17.2 Five mil click mechanism.- Set the lead knob in the DIRECT fire position and rotate the azimuth knob through the full range of operation. During rotation, a tactile examination shall be made to feel the detent action required in 5 mil increments. An audible examination shall be simultaneously performed to listen for a distinct click resulting from positive detent seating. At at least five readings throughout the range of operation, the displacement from one increment setting to the next shall represent a 5 mil angular change within the tolerance specified in 3.8.2.6. All the above examinations shall be performed at standard ambient temperature (+60° +90°F) and the extreme operating temperatures of 3.3.2 to determine compliance with 3.8.2.6.

4.6.18 15 Mils steps .- The telescope shall be set at 3200 (zero) on the azimuth counter and on the reset counter, with the gunner's aid counter set at zero. Rotate the azimuth knob to remove backlash. The vertical and horizontal lines of the telescope reticle shall be coincident with the target lines. Rotate the gunner's aid knob in the opposite direction to the rotation of the azimuth knob, and introduce 15 mils on the correct gunner's aid counter. Rotate the azimuth knob 15 mils in the same direction used to remove the backlash. The reset counter shall return to the original setting within the tolerance specified in 3.8.2.2. This test shall be repeated at each position indicated for a plus deflection and a minus deflection reading.

4.6.19 Backlash (azimuth).- The azimuth knob shall first be rotated in one direction to bring the vertical line of the telescope reticle in coincidence with the azimuth zero point on the target. Read and record the vernier scale readings on the azimuth and the reset counter scales. continue the rotation of the knob in the same direction, then reverse the direction of the knob and, while sighting through the telescope, reposition the vertical reticle line of the telescope in coincidence with the azimuth zero point on the target, without overtravel. Read the vernier scales again and record the readings, then compare them with the original set of readings. Any difference is the amount of backlash in the azimuth mechanism. The backlash shall not exceed the tolerances specified in 3.8.2.3.

4.6.20 Lift,- The vertical and horizontal lines on the telescope reticle shall be positioned on the target as in the initial position of 4.6.15. The azimuth knob shall be rotated, first in oone direction and then in the reverse direction. Lift is indicted if the line of sight is vertically displaced from the original zero elevation setting, and it shall not exceed the tolerance specified in 3.8.2.5. This examination shall be repeated at the 3200, 4800, zero, and 1600 mil settings.

4.6.21 Rapid traverse.- The azimuth knob of the telescope shall be rotated at the speed and for the length of time specified in 3.8.3.1. During this test, the reset counter shall follow the azimuth counter within the tolerance specified in 3.8.3.1. This test shall be conducted in both directions.

4.6.22 Reset counter.- The reset counter shall be resettable to 3200 when the knob is depressed and rotated in a counterclockwise direction, to determine compliance with 3.8.3.2.

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4.6.23 Gunner's aid counters.

4.6.23.1 Excursion range.- The gunner's aid counter shall be manually rotated **to** check a minimum of 50 mils excursion, as indicated in the right and left counter windows, to determine compliance with 3.8.3.3.1.

4.6.23,2 Settings.- Rotate the gunner's aid counter knob in the direction that will introduce a value on the "L" counter of from 1 mil to 50 mils from the zero setting. This will decrease the reset counter reading by the same value selected on the gunner's aid counter ±0.5 mil. Rotation in the opposite direction from zero to 1 mil through 50 mils will increase the reset counter reading by the value selected ±0.5 mil. Any directional change in the azimuth counter when the various values are put into the gunner's aid counter shall be within the limits specified in 3.8.3.3.2.

4.6.24 Operability.- The following components shall be operable at the extreme operating temperatures of 3.3.2 to conform to requirements of 3.9. The boresight adjustment shaft shall be manually operable with the aid of a screw driver. The azimuth door (counter cover) shall snap open when the latch lever is pushed back. The latch shall snap to secure the door when it is manually closed.

4.6.24.1 Torque.- The running torque required to rotate the knobs specified in 3.9.1, at standard ambient and extreme operating temperatures, shall be determined with a standard torque measuring device. The torque device shall be equipped with an adapter to fit the respective knob being tested. The measured torque values shall be within the limits specified in 3.9.1.

4.6.25 Direct force.

4.6.25.1 Elbow assembly.- A standard force measuring device shall be used for this test.The force required to rotate the elbow assembly in both directions shall not exceed the values specified in 3.9.2.1 when applied as directed.

4.6.25.2 Reset counter knob.- The direct force necessary to push in the knob to engage the reset counter shall be measured by means of standard measuring equipment that will measure applied force. The force applied shall not exceed the values specified for standard ambient temperature and extreme operating temperatures, to determine compliance with 3.9.2.2.

4.6.26 Illumination.- The counter illumination lamp shall be energized through the contact with 28 volts dc ±4 volts. The voltage shall be controlled through a voltmeter by means of a variac or potentiometer. With the light source energized and the telescope in **a** darkened area, the numerals and graduations on all counters shall be clearly visible, as required by 3.10.1.

4.6.26.1 Reticle.- A light source of 28 volts dc, 0.04 amperes, shall be positioned in the light port of the elbow assembly by means of an adapter (Figure 1). With the source energized and the telescope in a darkened area,

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the graduations and markings on the reticle shall be clear, and there shall be no flare or bright areas as the head is rotated one full turn in azimuth. The elbow assembly shall be rotated from one stop to the other, and it must meet the above conditions as required by 3.10.2.

4.6.27 Insulation resistance.- This test shall be conducted in accordance with Method 302 of Standard MIL-STD-202, using Test Condition "A". Insulation resistance measurements shall be made between insulated points and ground. Insulation resistance values shall not exceed the values specified in 3.11.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 8587340. The level of protection shall be as specified in procurement document.

6. NOTES

6.1 intended use.- The Panoramic Telescope M115, when mounted in Telescope Mount M137, is primarily intended to lay the 8 inch Self-Propelled Howitzer M11O and the 175MM Self-Propelled Gun M107 in azimuth for indirect fire.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of an applicable level of preservation, packaging, and packing.
(c) Applicable packaging data sheet number (see 5.1).
(d) Applicable stock number.

Custodian:

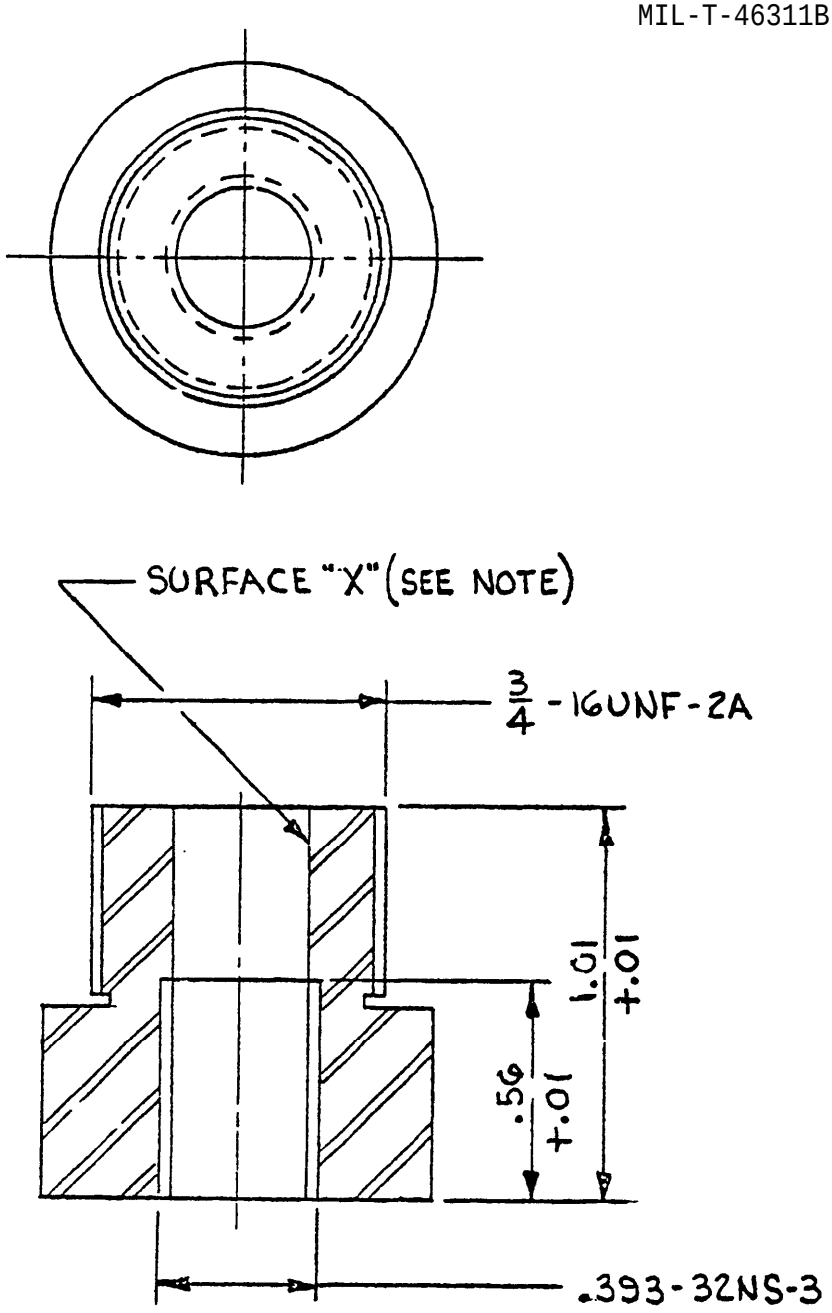
Army - MU

Preparing activity:

Army - MU

Project No. 1240-A186

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NOTE:- SURFACE "X" TO BE PAINTED WHITE

ADAPTER FOR RETICLE ILLUMINATION TEST

FIGURE #1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES ; NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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